Machine Learning: Midterm

Name: _____

1. Evaluating classifiers (20 points)

A course runs an AI-text detector on all 400 essays submitted for an assignment. An honor-code review later established that 60 of the essays were written with AI. The detector flagged 50 essays, and 45 of the flagged essays were written with AI. Treat "written with AI" as the positive class.

1.1 [4 points] Write down the confusion matrix.

1.2 [2 points] What are precision and recall for this detector?

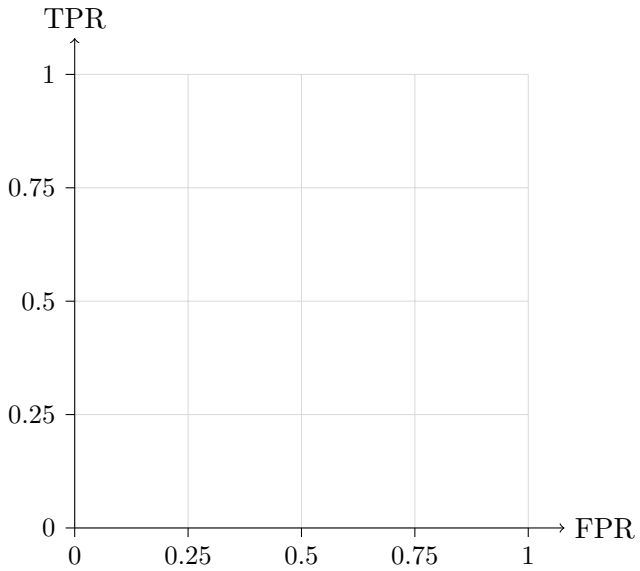
1.3 [2 points] Do we care more about improving precision or recall for this problem, and why?

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

essay	1	2	3	4	5	6	7	8
score	0.95	0.85	0.70	0.60	0.50	0.40	0.30	0.10
written with AI?	yes	yes	no	yes	no	yes	no	no

We say the detector flags an essay as AI-written if its score is over t.

1.4 [10 points] Draw the ROC curve.



1.5 [2 points] Compute the AUC.

2. Support vector machines (20 points)

2.1 [10 points] A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

point	A	B	C	D	E
x_1	3	1	1.5	1.25	3
x_2	1	1	1.5	1.5	0.5
label	+	−	+	−	−

(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.

(b) What is the width of the margin?

(c) Compute each point's distance to the decision boundary.

(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?

2.2 [4 points] The soft-margin SVM solves

$$\min_{\boldsymbol{w},\, b,\, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w}\cdot\boldsymbol{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

(a) What does the slack ξ_i measure, and why do we need it?

(b) Where does a training point lie, relative to the margin and the decision boundary, if $\xi_i = 0$? If $0 < \xi_i < 1$? If $\xi_i > 1$?

2.3 [4 points] Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$.

(a) Which has the wider margin, and why?

(b) Which has more support vectors, and why?

(c) Which is more likely to overfit, and why?

(d) How would you choose C in practice, and why?

2.4 [2 points] You remove a training point that is not a support vector and retrain the SVM. What changes? Why?